

Yichen Song

Internship Availability: Flexible

<https://yichenso.github.io/>

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Research Interests

I am interested in developing multi-task and continual-learning agents. To this end, I study reinforcement learning and (nonstationary) bandits in both online and offline settings. Recently, I have been particularly excited about representation learning in RL, as I believe that learning effective representations is key to building such agents.

Education

Boston University

Ph.D. in Computing and Data Science (GPA: 3.95/4.0)

Advisor: Aldo Pacchiano

Boston, MA

Sept. 2023 – Present

University of Michigan - Ann Arbor

Master of Science in Robotics (GPA: 3.95/4.0)

Ann Arbor, MI

Sept. 2021 – Apr. 2023

Southern University of Science and Technology (SUSTech)

Bachelor of Engineering in Robotics (GPA: 3.85/4.0)

Shenzhen, China

Sept. 2017 – Jun. 2021

Work Experience

Honda Research Institute USA, Inc.

Research Intern | Mentor: Aolin Xu

San Jose, CA

May 2022 – Aug. 2022

- Proposed a graph-based surrounding vehicle trajectory prediction framework based on various vehicle behavior models.
- Implemented the proposed framework in the CARLA simulator and developed an associated warning generation system for the ego vehicle based on the prediction results.

Publications

- [1] **Y. Song**, A. Russo, A. Pacchiano, “Future Information-Directed Sampling for Bayesian Nonstationary Bandits”, *ICML Workshop on Decision-Making from Offline Datasets to Online Adaptation: Black-Box Optimization to Reinforcement Learning*, 2026.
- Developed Future Information-Directed Sampling (FIDS), a future-aware algorithm for Bayesian nonstationary bandits that leverages predictive information structures, with regret guarantees comparable to Thompson Sampling and an offline supervised-learning framework that uses Transformer in-context learning to approximate FIDS policies from offline data.
- [2] A. Russo, **Y. Song**, A. Pacchiano, “Pure Exploration with Feedback Graphs”, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025, **Oral presentation**
- Developed a theoretical framework for pure exploration with feedback graphs, encompassing feedback structures from full information and standard bandit feedback to settings without self-observations; derived instance-dependent sample-complexity lower bounds and proposed TaS-FG, an asymptotically optimal best-action identification algorithm.
- [3] H. Zhou, **Y. Song**, V. Tzoumas, “Safe Non-Stochastic Control of Control-Affine Systems: An Online Convex Optimization Approach”, *IEEE Robotics and Automation Letters*, 2023
- Developed a safe online control framework for nonlinear control-affine systems under bounded non-stochastic disturbances and time-varying convex safety constraints, with bounded dynamic regret against a clairvoyant controller, and validated it on inverted-pendulum stabilization and quadrotor navigation tasks.

Professional Activities and Service

Reviewers for AISTATS (2026), ICLR (2025, 2026), Neurips (2025, 2026)

Teaching Experience

Teaching Fellow, Boston University

- Algorithms for Data Science (Fall 2025)
- Introduction to Sequential Decision Making (Spring 2026)

Skills

Languages: Python, C++, MATLAB

Frameworks/Tools: PyTorch, TensorFlow, ROS